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• **Description of model architecture and parameter count.**

The model consists of a deep convolutional neural network trained on peptide data derived from the NIST peptide spectral library using approximately 1,721,730 parameters from the human dataset. The model's goal is to predict fragment ion intensities (for sequences of 5 to 30 amino acids in length with CAM/Cysteine modifications or no modifications only).

Each peptide sequence is first converted into 128 dimension learned embedding representation (using 20 standard amino acids + 1 padding token). The embedded sequence is then processed through the convolutional layers with different kernel sizes (2,3,5,7). The outputs from these layers are then concatenated and goes through a stack of 4 residual blocks. Then the model incorporates dropout rate of 0.2 after the residual layers which reduces overfitting. Finally, the network produces predictions for each b/y ions using a linear output layer, with a sigmoid activation function applied to it.

• **Data preprocessing details and normalization procedure.**

The data is parsed/extracted with matchms library in python. The sequences are filtered according to the rules stated by the assignment. (length of 5 to 30, CAM modifications with C only, other AAs unmodified). For each intensity and b/y ions, only ones with charge = +1 are used. Multiple annotations per peak were handled by matching nearest m/z within the tolerance of 1e-4. The intensities are then normalized with its max intensity so that all intensities are between 0 and 1. Also, any value below the threshold of < 1e-2 is discarded/set to 0.

• **Performance metrics on held-out data from the training dataset.**

The performance metrics for the human dataset is as follows: (output from the code training file)

Loaded 73222 spectra
Parameters: 1721730
Epoch 1: loss=0.055507, MSE=0.027753, binary_acc=0.7430
Epoch 2: loss=0.043615, MSE=0.021807, binary_acc=0.7693
Epoch 3: loss=0.041307, MSE=0.020654, binary_acc=0.7809
Epoch 4: loss=0.039941, MSE=0.019970, binary_acc=0.7858
Epoch 5: loss=0.038902, MSE=0.019451, binary_acc=0.7885
Epoch 6: loss=0.038036, MSE=0.019018, binary_acc=0.7915
Epoch 7: loss=0.037273, MSE=0.018637, binary_acc=0.7930
Epoch 8: loss=0.036565, MSE=0.018283, binary_acc=0.7948
Epoch 9: loss=0.035873, MSE=0.017936, binary_acc=0.7951
Epoch 10: loss=0.035194, MSE=0.017597, binary_acc=0.7978
Epoch 11: loss=0.034599, MSE=0.017299, binary_acc=0.7984
Epoch 12: loss=0.034015, MSE=0.017008, binary_acc=0.8011
Epoch 13: loss=0.033516, MSE=0.016758, binary_acc=0.8019
Epoch 14: loss=0.033036, MSE=0.016518, binary_acc=0.8028
Epoch 15: loss=0.032417, MSE=0.016208, binary_acc=0.8030
Epoch 16: loss=0.031988, MSE=0.015994, binary_acc=0.8051
Epoch 17: loss=0.031563, MSE=0.015781, binary_acc=0.8050
Epoch 18: loss=0.031153, MSE=0.015577, binary_acc=0.8056
Epoch 19: loss=0.030757, MSE=0.015378, binary_acc=0.8068
Epoch 20: loss=0.030405, MSE=0.015203, binary_acc=0.8080

• **Figures that illustrate the comparison between predicted and experimental spectra for 4 canonical peptides, including peptide, charge state, and full MSP Name line.**

Figure 1. Spectrum Similarity comparison for sequence MSP line name = 'FANSLVGVQQQLQAFNTYR/3_0'

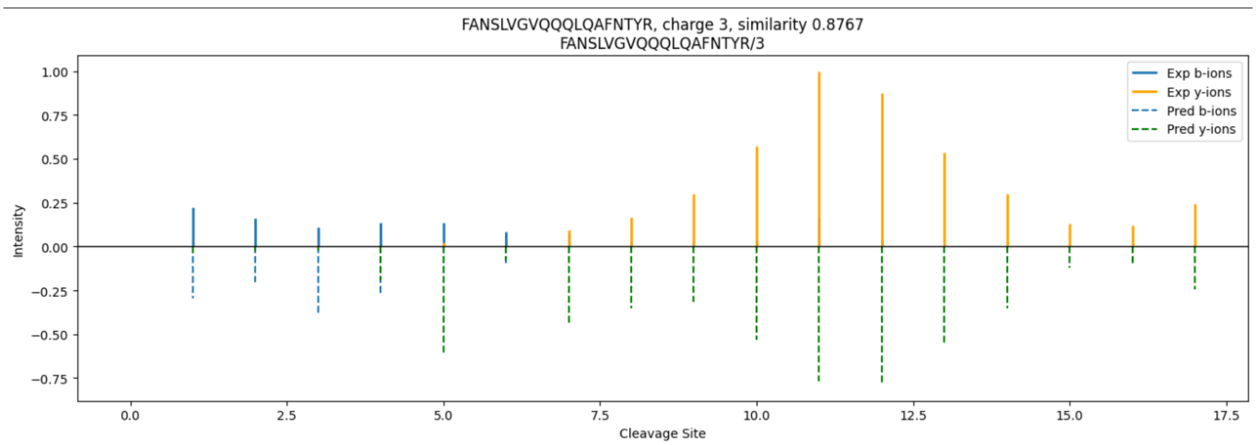

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Figure 2. Spectrum Similarity comparison for sequence MSP line name = 'AEKEEENR/2_0'

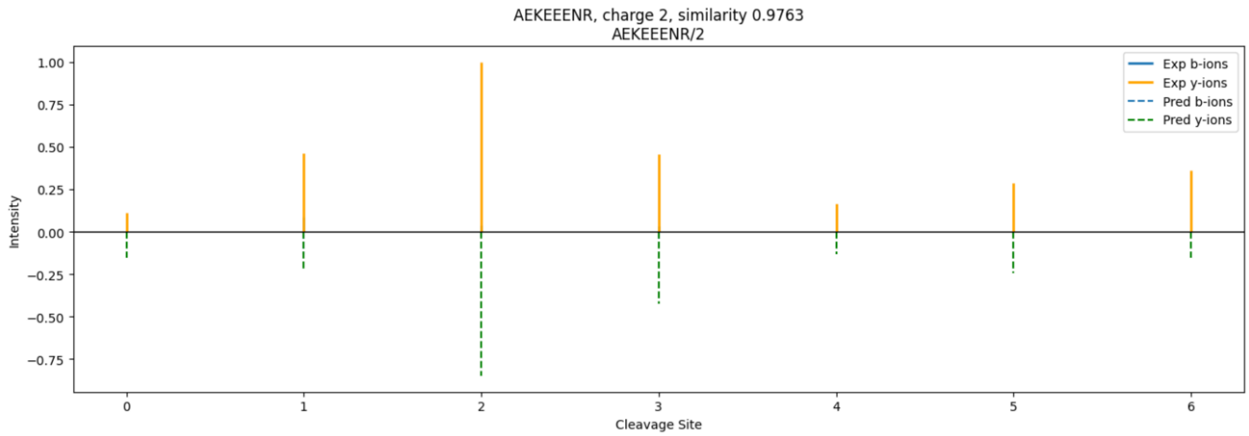

Figure 3. Spectrum Similarity comparison for sequence MSP line name = 'EQMDQISESNHLIHLSNTNNVK/3_0'

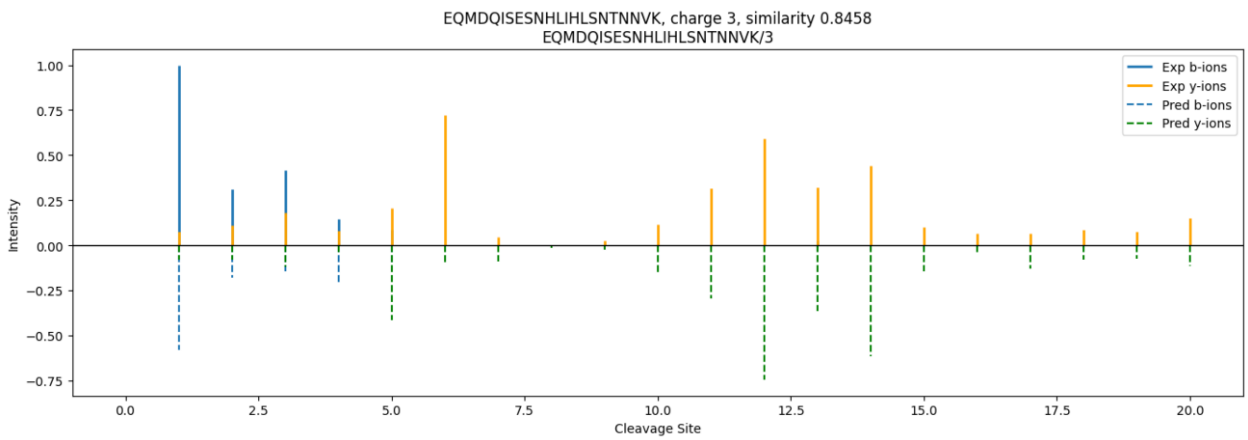

Figure 4. Spectrum Similarity comparison for sequence MSP line name = 'AEKPPLSASSPQQRPPEPETGESAGTSR/3_0'